

AKQoL Item Fit Analysis

Empirical critical values for polytomous Rasch item fit statistics

18 september 2026

1 Data and method

The AKQoL (actinic keratosis quality of life) questionnaire consists of 10 items (Q1-Q10), originally scored 0-3. Following a check of category usage, categories 2 and 3 were collapsed for all 10 items, giving a 0-2 scale. 3 respondent(s) with an extreme (minimum or maximum possible) total score were excluded, since their person-parameter estimate is undefined, leaving $N = 96$ respondents.

Item and person parameters were estimated by conditional maximum likelihood (CML, via `eRm::PCM`) and weighted maximum likelihood (WML) respectively, using the `RASCHplot` package (v0.1.0) with the local patch (`RASCHplot_patches.R`) applied — this fixes an item/ threshold orientation bug in `pcmfmt()`/`rRMPstats()` and a numerical-overflow issue that otherwise crashes the simulation for polytomous data. The empirical sampling distribution of the minimum and maximum item fit statistics across all 10 items was estimated from $B = 300$ parametric bootstrap replicates (`rRMPstats()`), refitting CML/WML on each simulated data set.

2 Item parameters

Table 1: Estimated item step (threshold) parameters (CML).

Item	Threshold 1	Threshold 2
Q1	-0.508	0.289
Q2	-0.160	-0.725
Q3	1.388	3.988
Q4	-1.948	-4.631
Q5	3.348	4.766
Q6	-0.911	0.649
Q7	-2.752	-5.395
Q8	2.154	4.021
Q9	-3.653	-7.028
Q10	0.848	0.420

3 Evaluation of item fit

Observed fit statistics for each item, compared against the empirical null distribution of the minimum/maximum across items from the 300 simulated replicates. * marks an item beyond the 5%/95% empirical percentile, ** beyond the 2.5%/97.5% percentile (two-sided, unadjusted for multiple testing).

Table 2: Observed item fit statistics, flagged against the empirical null. Outfit is not flagged against a maximum threshold, and Infit not against a minimum threshold, since those reference distributions are degenerate (see Section 4).

Item	Outfit	Infit	tOutfit	tInfit	FitResid
Q1	1.322	0.907	1.933	-0.806	2.303
Q2	0.960	0.735	-0.102	-2.523	0.339
Q3	1.374	1.601	1.451	3.114	1.603
Q4	43.451	22.514	5.111	8.969	2.073
Q5	7.388	7.406	4.342	7.315	2.708
Q6	1.676	1.732	4.069	5.106	4.046
Q7	131.866	62.530	5.950	8.127	1.853
Q8	3.374	3.574	4.131	6.815	3.160
Q9	317.481	161.438	5.077	5.374	0.987
Q10	3.330	2.848	7.019	9.634	5.498

4 Simulated empirical null distributions

The minimal-Outfit and maximal-Infit distributions are both degenerate under this model, for the same reason: in every one of the 300 replicates, at least one item has zero total weighted residual variance in the simulated data set, giving that item $\text{Outfit} = 0$ and $\text{Infit} = \infty$ simultaneously. This is not a fixed item - it rotates among Q3, Q4, Q5, Q7, Q8 and Q9 (the same items flagged in Table 2), confirming that their extreme observed fit reflects a genuinely fragile CML calibration for this collapsed scale rather than a one-off artifact. Both are shown as such below rather than as a (misleading) density plot. The maximal-Outfit distribution is not degenerate, but its right tail is extreme (up to the order of 10^{90} in a handful of replicates, again traceable to the same fragile items being refit on resampled data), so it is plotted on a $\log_{10}$ x-axis; even so, its median (928) is far above what an Outfit statistic should be under good fit, so this null distribution should not be treated as a usable empirical critical value without first addressing why Q3-Q9 refit so unstably.

Minimal Outfit

min Outfit is = 0
in all 300 replicates.
Item achieving it:
Q5 (132/300), Q4
(73/300), Q3 (46/300),
Q7 (34/300), Q9
(9/300), Q8 (6/300)

Maximal Outfit

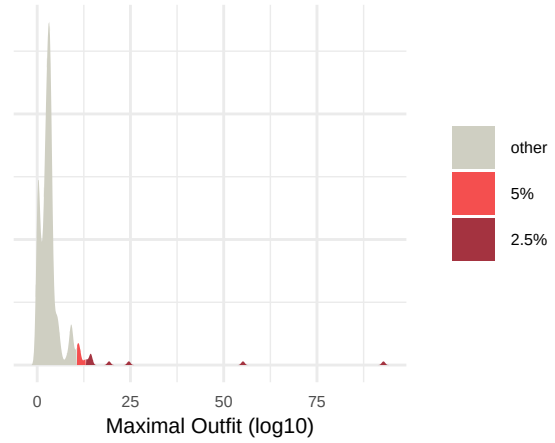


Figure 1: Empirical null distribution of the minimum and maximum Outfit across items.

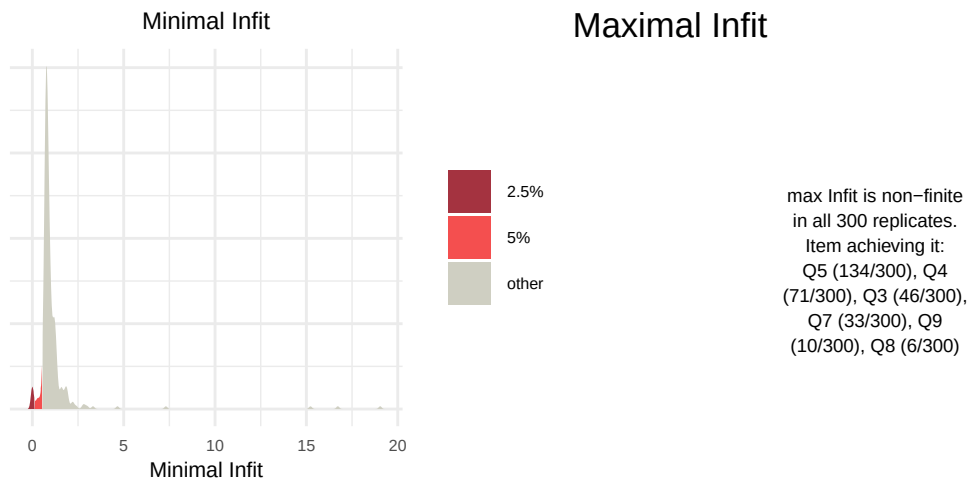


Figure 2: Empirical null distribution of the minimum and maximum Infit across items.

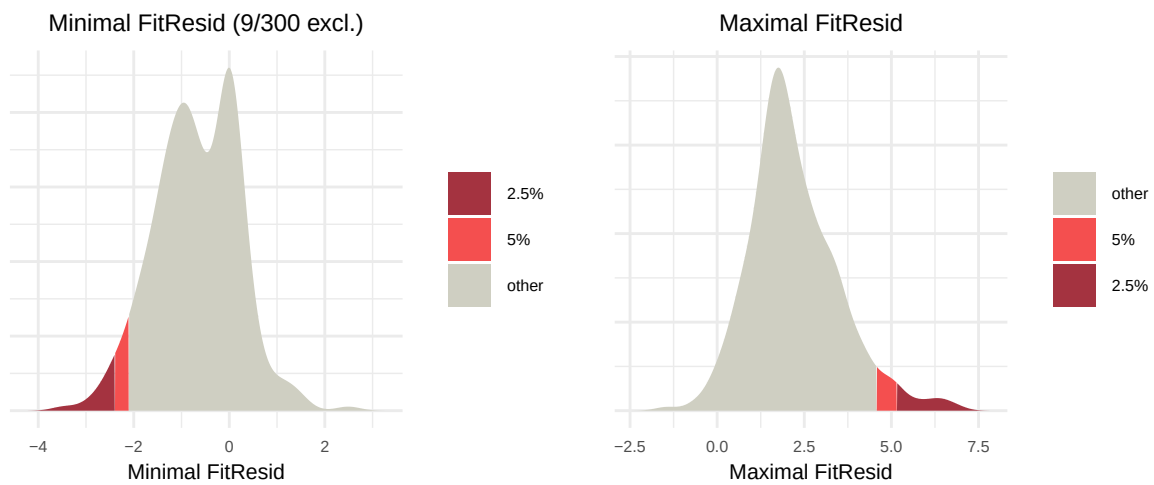


Figure 3: Empirical null distribution of the minimum and maximum FitResid across items.